The Poisson Distribution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

Events occur independently of one another; events occur at a constant average rate over the given interval (of time or space); two events cannot occur at exactly the same instant; the number of events in non-overlapping intervals is independent.

2 **[Foundational] • Target: Computing Poisson probabilities**

A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, to three decimal places.

$P(X = 6) = e^{-4}4^6/6! = (0.0183)(4096)/720 \approx 74.98/720 \approx 0.104$.

3 **[Foundational] • Target: Computing Poisson probabilities**

Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.

$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2)$. $P(X = 0) = e^{-4} \approx 0.0183$. $P(X = 1) = 4e^{-4} \approx 0.0733$. $P(X = 2) = 16e^{-4}/2 = 8e^{-4} \approx 0.1465$. Sum $\approx 0.0183 + 0.0733 + 0.1465 \approx 0.238$.

4 **[Foundational] • Target: Mean and variance of a Poisson distribution**

State the mean and variance of a Poisson distribution with parameter λ, and explain what is distinctive about this relationship compared to other common distributions.

For a Poisson distribution, both the mean and the variance equal λ: $E(X) = \lambda$ and $\text{Var}(X) = \lambda$. This is distinctive because for most other distributions, the mean and variance are independent parameters (e.g. the normal distribution has separate μ and σ^2) — for Poisson, a single parameter determines both.

5 **[Foundational] • Target: Sums of independent Poisson variables**

Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.

The sum of independent Poisson random variables is itself Poisson: $X_1 + X_2 \sim \text{Po}(3 + 5) = \text{Po}(8)$. Mean $= 8$; variance $= 8$ (since for Poisson, mean and variance are equal).

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.

The Poisson approximation to the binomial is appropriate when n is large and p is small (a common guideline: $n > 50$ and $np < 5$), so that events are rare relative to the number of trials. In that case, X is approximately $\text{Po}(\lambda)$ with $\lambda = np$.

7 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.

$\lambda = np = 2000(0.002) = 4$. $P(X = 3) = e^{-4}4^3/3! = (0.0183)(64)/6 \approx 1.171/6 \approx 0.195$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using cumulative Poisson probabilities

A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.

$P(X > 2) = 1 - P(X \leq 2)$. Summing the three given probabilities:
$P(X \leq 2) = 0.0067 + 0.0337 + 0.0842 = 0.1246$. So $P(X > 2) = 1 - 0.1246 = 0.8754$.

9 [Intermediate] • Target: Finding an unknown parameter from a given probability

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ, to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.

$e^{-\lambda} = 0.1353$. Taking natural logs: $-\lambda = \ln(0.1353) \approx -2.0$, so $\lambda \approx 2$.

10 [Intermediate] • Target: Variance-based reasoning with the Poisson distribution

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 3. Find λ, and then find $P(X = \lambda)$, to three decimal places.

Since variance $= \lambda$ and standard deviation $= \sqrt{\lambda} = 3$, we get $\lambda = 9$.

$P(X = 9) = e^{-9}9^9/9! = (0.0001234)(387420489)/362880 \approx 47790/362880 \approx 0.132$.

© ClipSAT — clipsat.org | All Rights Reserved

11 **[Intermediate]** • **Target: Combining Poisson reasoning with real-world decisions**

A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.6065 = 0.3935$, or about 39.35%. Since this exceeds their target threshold of 30%, the team should be concerned — there is roughly a 39% chance of at least one crash in a given week, above their comfort threshold, suggesting they should investigate ways to reduce the average crash rate λ.

12 **[Intermediate]** • **Target: Combining a Poisson approximation with conditional probability**

Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.

$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.2231 = 0.7769$. Conditional probability:

$P(X = 1 \mid X \geq 1) = P(X = 1)/P(X \geq 1) = 0.3347/0.7769 \approx 0.431$.

13 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

For a 3-second interval, the parameter becomes $\lambda = 2 \times 3 = 6$.

$P(X = 3) = e^{-6}6^3/3! = (0.002479)(216)/6 \approx 0.5354/6 \approx 0.089$.

14 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.

The number of heads in 10 coin flips is naturally modeled by a binomial distribution $B(10, 0.5)$, not a Poisson distribution, because a Poisson model requires events to be rare (a small probability of occurrence per trial, with np moderate while n is large) — here $p = 0.5$ is far too large for the Poisson approximation to be valid, and there is a fixed, small number of trials ($n = 10$) rather than events occurring continuously over time or space.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

15 [Challenging] • Target: Comparing two independent Poisson processes • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.

Since the two processes are independent, the joint probability is the product of the individual probabilities: $P(X_A = 2) \times P(X_B = 1) = 0.2240 \times 0.2707 \approx 0.0606$.

16 [Challenging] • Target: Rescaling the Poisson parameter for a different interval • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A radio station receives an average of 12 requests per hour. Find the appropriate Poisson parameter λ to use for a 15-minute segment.

Since the rate is 12 per hour and 15 minutes is a quarter of an hour, the parameter for that segment is $\lambda = 12/4 = 3$.

17 [Challenging] • Target: The mode of a Poisson distribution • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

A random variable $X \sim \text{Po}(4)$ has an integer parameter $\lambda = 4$. State the mode(s) of X (the most likely value or values), and explain why a Poisson distribution with an integer λ has two modes, using the ratio $P(X = k)/P(X = k - 1) = \lambda/k$.

At $k = \lambda$, the ratio $P(X = k)/P(X = k - 1) = \lambda/k = \lambda/\lambda = 1$, so $P(X = \lambda) = P(X = \lambda - 1)$ exactly. This means both $X = 3$ and $X = 4$ share the same (maximum) probability, so the distribution has two modes: 3 and 4. This only happens when λ is a positive integer; for a non-integer λ, there is a single mode at $\lfloor \lambda \rfloor$.

18 [Challenging] • Target: Assessing Poisson plausibility from sample data • Cross-Lesson: 5.01 Further Integration Techniques

A biologist counts the number of bacteria colonies in 10 equal-sized petri dishes: $3, 5, 4, 6, 5, 4, 3, 5, 6, 4$. Find the sample mean and the unbiased sample variance, and comment on whether a Poisson model seems plausible for this data.

Mean $= (3 + 5 + 4 + 6 + 5 + 4 + 3 + 5 + 6 + 4)/10 = 45/10 = 4.5$. Squared deviations from the mean sum to $4(2.25) + 6(0.25) = 9 + 1.5 = 10.5$ (the values 3 and 6 each deviate by 1.5, contributing 2.25 each, four times in total; the values 4 and 5 each deviate by 0.5, contributing 0.25 each, six times in total). Unbiased sample variance: $10.5/(10 - 1) = 10.5/9 \approx 1.167$. Since the sample variance (≈ 1.17) is much smaller than the sample mean (4.5), a Poisson model seems implausible for this data — a genuine Poisson dist.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 5.02 Differential Equations

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

(a) In a 2-week period, the parameter scales to $\lambda = 1.6 \times 2 = 3.2$:

$P(X \geq 3) = 1 - P(X \leq 2) = 1 - e^{-3.2}\left(1 + 3.2 + \frac{3.2^2}{2}\right) = 1 - 9.32e^{-3.2} \approx 1 - 0.3799 = 0.620$ (3 s. f.) (b) For 1 week, $\lambda = 1.6$. Use conditional probability:

$P(X = 2 \mid X \geq 1) = \frac{P(X=2)}{P(X \geq 1)} = \frac{\frac{e^{-1.6}(1.6)^2}{2}}{1 - e^{-1.6}} = \frac{1.28e^{-1.6}}{1 - e^{-1.6}} \approx \frac{0.2584}{0.7981} \approx 0.324$ (3 s. f.)

20 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

Let $X \sim B(12, 0.05)$. We need $P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$. $P(X = 0) = (0.95)^{12} \approx 0.5404$. $P(X = 1) = 12(0.05)(0.95)^{11} \approx 0.3413$. Thus, $P(X \geq 2) = 1 - (0.5404 + 0.3413) = 1 - 0.8817 = 0.118$ (3 s.f.).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying an appropriate Poisson scenario

Which of the following scenarios is best modeled by a Poisson distribution?

a. The number of typing errors on a randomly selected page of a manuscript. ✓

b. The number of heads in 20 coin tosses.

c. The height of a randomly selected adult.

d. The number of correct answers on a 10-question multiple-choice test, each with 4 options.

Typing errors occur rarely and independently across a fixed amount of text (a page), matching the Poisson conditions of a constant average rate with no fixed number of trials. The coin-toss and multiple-choice scenarios are binomial (a fixed number of trials, each with the same success probability), and height is a continuous variable following approximately a normal distribution — neither fits the Poisson framework.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Computing Poisson probabilities

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

a. **0.980** ✓

b. 0.900

c. 0.810

d. 0.990

1. **Margin of error for each confidence interval:** For a symmetric 90% confidence interval with known σ, the critical value is $z = 1.645$. Each interval has half-width: $E = 1.645\frac{\sigma}{\sqrt{n}}$ The two intervals are $I_1 = [\bar{X}_1 - E, \bar{X}_1 + E]$ and $I_2 = [\bar{X}_2 - E, \bar{X}_2 + E]$. 2. **Condition for interval overlap:** Two intervals of identical half-width E overlap if and only if the distance between their centres does not exceed the sum of their half-widths.

23 [Foundational] • Target: Mean and variance of a Poisson distribution

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

a. $\frac{19}{27}$ ✓

b. $\frac{8}{27}$

c. $\frac{7}{8}$

d. $\frac{2}{3}$

Let $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(2\lambda)$ be independent Poisson random variables. The sum $T = X + Y$ follows a Poisson distribution with parameter $\lambda + 2\lambda = 3\lambda$. The conditional distribution of X given $X + Y = n$ is binomial $\mathrm{B}\left(n, \frac{\lambda}{\lambda + 2\lambda}\right) = \mathrm{B}\left(n, \frac{1}{3}\right)$. Here, $n = 3$, so $X \mid (X + Y = 3) \sim \mathrm{B}\left(3, \frac{1}{3}\right)$. We want to find.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Sums of independent Poisson variables

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests: $H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$ The two-stage procedure operates as follows: - **Stage 1:** A $1-\text{metre}$ sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \geq 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2. - **Stage 2:** A second, independent $1-\text{metre}$ sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \geq 4$, H_0 is rejected; otherwise, H_0 is accepted. Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

a. $e^{-2} + 16e^{-4} \approx 0.428$ ✓

b. $16e^{-4} \approx 0.293$

c. $4e^{-2} \approx 0.541$

d. $1 - e^{-1} - \frac{7}{2}e^{-2} \approx 0.158$

A Type II error occurs when H_0 is accepted despite the true parameter value being $\lambda = 2$ (i.e. H_1 is true). Under $\lambda = 2$, the independent counts $X_1, X_2 \sim \text{Po}(2)$. The probability mass function is: $P(X = k) = \frac{2^k e^{-2}}{k!}$ Specifically: $P(X = 0) = e^{-2}, \quad P(X = 1) = 2e^{-2}, \quad P(X = 2) = 2e^{-2}$ H_0 can be accepted in two mutually exclusive ways: 1. **In Stage 1:** H_0 is accepted immediately if $X_1 = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. **0.08 ✓**

c. 0.12

d. 0.16

Part (i): For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$. \\ We are given $n = 25$ and $E(X) = 2$. \\ So, $25p = 2$, which gives $p = \frac{2}{25} = 0.08$. \\ Part (ii): Now, $n = 50$ and $p = 0.08$. The number of defective bulbs follows $X \sim B(50, 0.08)$. \\ We need to find the smallest integer k such that $P(X \leq k) > 0.9$. \\ We can use a binomial cumulative distribution table or a calculator. \\ Let's calculate a few values: $P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$ Using.

26 [Foundational] • Target: Using cumulative Poisson probabilities

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. **44 ✓**

b. 43

c. 11

d. 22

We solve the problem in two stages: **Step 1: Determine σ** Under H_0, the distribution of the sample mean is $\bar{X} \sim N\left(20.0, \frac{\sigma^2}{25}\right)$. The critical value for a upper one-tailed test at the 5% level corresponds to $z = 1.645$: $\frac{20.658 - 20.0}{\frac{\sigma}{\sqrt{25}}} = 1.645$ $\frac{0.658}{\frac{\sigma}{5}} = 1.645 \Rightarrow \frac{3.29}{\sigma} = 1.645 \Rightarrow \sigma = \frac{3.29}{1.645} = 2.0$ mm **Step 2: Determine the minimum sample size m** For.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Finding an unknown parameter from a given probability

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

a. The machine is likely calibrated correctly, as μ_0 is within the interval.

b. **The machine is likely not calibrated correctly, as μ_0 is outside the interval.** ✓

c. The machine is likely calibrated correctly, as the sample mean is close to μ_0.

d. The machine is likely not calibrated correctly, as the sample standard deviation is too large.

A) The standard error of the mean (SE) is calculated using the formula: $SE = \frac{s}{\sqrt{n}}$, where s is the sample standard deviation and n is the sample size. Given $s = 3.5$ grams and $n = 50$. $SE = \frac{3.5}{\sqrt{50}} \approx \frac{3.5}{7.071} \approx 0.495$ grams. B) To construct a 95% confidence interval for the mean weight, we use the formula: $CI = \bar{x} \pm z^* \times SE$, where $\bar{x}$ is the sample mean and z^* is the critical z-value for a 95% confidence level. For a 95% confidence lev.

28 [Foundational] • Target: Variance-based reasoning with the Poisson distribution

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield ≥ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

a. 1/2

b. **5/8** ✓

c. 3/8

d. 1/4

Let Y be the event that the yield is less than 50 kg, and B be the event that the fertilizer type is B. We are asked to find the conditional probability P(B|Y). From the table, we have: Number of plots with yield < 50 kg (event Y) = 80. Number of plots with yield < 50 kg AND fertilizer type B (event B ∩ Y) = 50. The formula for conditional probability is P(B|Y) = P(B ∩ Y) / P(Y). Alternatively, using the counts directly: P(B|Y) = (Number of plots with yield < 50 kg and fertilizer type B) / (Number of plot.

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Combining Poisson reasoning with real-world decisions**

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$ ✓

b. $B(6, 1/4)$

c. $B(1, 4/6)$

d. $B(4, 4/6)$

The problem describes a situation with a fixed number of independent trials (rolling a die 4 times), where each trial has only two possible outcomes (obtaining a 6 or not obtaining a 6), and the probability of success (obtaining a 6) is constant for each trial. In a binomial distribution, denoted as $X \sim B(n, p)$, n represents the number of trials, and p represents the probability of success on a single trial. From the question: - The number of times the die is rolled is 4, so $n = 4$. - The probability of o.

30 **[Intermediate]** • **Target: Combining a Poisson approximation with conditional probability**

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

a. **Yes — fixed, independent trials with constant success probability** ✓

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Combining Poisson processes over different time intervals

A grouped frequency table has class intervals of width 10 starting at 12, with frequencies $[10, 4, 5, 6]$ respectively. Using midpoints, find the mean.

a. **149/5 ✓**

b. 144/5

c. 154/5

d. 159/5

Midpoints: $[17, 27, 37, 47]$. Mean $= \frac{\sum fx}{\sum f} = \frac{745}{25} = 149/5$

32 [Intermediate] • Target: Comparing two independent Poisson processes

$X \sim Po(3)$ and $Y \sim Po(3)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 9

b. 0

c. **6 ✓**

d. 7

Since X and Y are independent Poisson variables, $X + Y \sim Po(3 + 3) = Po(6)$.

33 [Intermediate] • Target: Rescaling the Poisson parameter for a different interval

A binomial random variable has $n = 200$, $p = \frac{1}{50}$, and is approximated by a Poisson distribution. Find the parameter λ of the approximating Poisson distribution.

a. 50

b. **4 ✓**

c. 200

d. 5

$\lambda = np = 200 \times \frac{1}{50} = 4$.

34 [Intermediate] • Target: The mode of a Poisson distribution

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \leq h < 70$ is 1.2, find the value of k.

a. 10

b. **12 ✓**

c. 15

d. 20

The area of each bar in a histogram represents the frequency of the corresponding class interval. The frequency density is given by $\frac{\text{frequency}}{\text{class width}}$. The class interval $60 \leq h < 70$ has a class width of $70 - 60 = 10$. The frequency density for this interval is given as 1.2. Therefore, the frequency for this interval is frequency density $\times$ class width $= 1.2 \times 10 = 12$. The histogram shows that the frequency for the interval $60 \leq h < 70$ is represented by k..

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Identifying an appropriate Poisson scenario

A discrete random variable has $P(X = 1) = k$, $P(X = 2) = 2k$, $P(X = 3) = 3k$. Find k.

a. $\frac{5}{12}$

b. $\frac{2}{3}$

c. $\frac{1}{6}$ ✓

d. $\frac{5}{6}$

Working through this statistics step: $k + 2k + 3k = 1 \Rightarrow 6k = 1 \Rightarrow k = \frac{1}{6}$

36 [Intermediate] • Target: Assessing Poisson plausibility from sample data

A random variable X equals 2 with probability $\frac{1}{5}$ and 4 with probability $\frac{4}{5}$. Find $\mathrm{Var}(X)$.

a. $\frac{2}{5}$

b. $\frac{16}{25}$ ✓

c. $\frac{33}{25}$

d. $\frac{39}{50}$

$E(X) = \frac{18}{5},\ E(X^2) = \frac{68}{5},\ \mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{16}{25}$

37 [Intermediate] • Target: Conditions for a Poisson distribution

If $X \sim \mathrm{Po}(4)$, find $P(X = 3)$, correct to 4 decimal places.

a. 0.2454

b. 0.1563

c. 0.1465

d. **0.1954** ✓

Working through this statistics step: $P(X = 3) = \frac{e^{-4}4^3}{3!} \approx 0.1954$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Computing Poisson probabilities • Cross-Lesson: 5.01 Further Integration Techniques

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour. Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

a. $54e^{-6} \approx 0.134$ ✓

b. $\frac{2}{3}e^{-2} \approx 0.0902$

c. $\frac{2187}{8}e^{-9} \approx 0.0337$

d. $\frac{625}{24}e^{-5} \approx 0.175$

1. **Find the parameter λ for a 1-hour period:** Let X be the number of calls received in 1 hour. Then $X \sim \text{Po}(\lambda)$. The probability mass function is $\text{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$. We are given: $\text{P}(X = 1) = 1.5\,\text{P}(X = 3)$ Substitute the formula for each probability.

39 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 5.02 Differential Equations

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.

a. **a = 1.5, b = -30** ✓

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$. We are given $\mu_X = 100$, $\sigma_X = 15$, $\mu_Y = 120$, and $\sigma_Y = 22.5$. The linear transformation is given by $Y = aX + b$. We know that for a linear transformation $Y = aX + b$, the mean of Y is $\mu_Y = a\mu_X + b$ and the standard deviation of Y is $\sigma_Y = |a|\sigma_X$. From the standard deviation relationship: $\sigma_Y = |a|\sigma_X$ $22.5 = |a| \times 15$ $|a| = \frac{22.5}{15} = 1.5$ Now consider the mean relationship.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

The unique real root α of the equation $x^3 + 2x - k = 0$, where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$ ✓

b. $0 < k < 2\sqrt{2}$

c. $0 < k < 3$

d. $0 < k < 12$

To determine the range of k for which the iteration converges locally, we follow these steps: 1. **Root relation:** At the fixed point $x = \alpha$, we have: $\alpha = g(\alpha) = \frac{k}{\alpha^2 + 2} \Rightarrow \alpha^3 + 2\alpha = k$. Since $k > 0$ and the function $f(x) = x^3 + 2x - k$ satisfies $f'(x) = 3x^2 + 2 > 0$ for all real x, there is a unique real root α, and $\alpha > 0$. 2. **Differentiating $g(x)$:** Using the chain rule on $g(x) = k(x^2 + 2)^{-1}$.

41 [Challenging] • Target: Poisson as an approximation to the binomial • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

a. $\frac{1}{8}$

b. $\frac{3}{8}$ ✓

c. $\frac{1}{2}$

d. $\frac{7}{8}$

This is a binomial distribution scenario. The number of trials is $n = 3$ (the number of coin tosses). The probability of success (getting a head) in a single trial is $p = \frac{1}{2}$, and the probability of failure (getting a tail) is $q = 1 - p = \frac{1}{2}$. We want to find the probability of getting exactly $k = 2$ heads. The binomial probability formula is $P(X = k) = \binom{n}{k} p^k q^{n-k}$. Substituting the values: $P(X = 2) = \binom{3}{2} (\frac{1}{2})^2 (\frac{1}{2})^{3-2}$.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Using cumulative Poisson probabilities • Cross-Lesson: 6.01 Matrices (AI HL)

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$**; reject** H_0 **because** $-2.0 < -1.645$ ✓

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

1. Under the null hypothesis $H_0 : \mu = 25.0$, the sample mean $\bar{X}$ of a random sample of size $n = 16$ from $\text{N}(\mu, \sigma^2)$ has standard error: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{16}} = \frac{0.8}{4} = 0.2$ 2. The standardized test statistic z is: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{24.6 - 25.0}{0.2} = \frac{-0.4}{0.2} = -2.0$ 3. Since this is a lower-tail test at the 5% level with critical value $z_{\text{crit}} = -1.645$, the critical (rejec.

43 [Challenging] • Target: Finding an unknown parameter from a given probability • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

a. **0.9545** ✓

b. 0.6827

c. 0.9772

d. 0.0228

Let X be the weight of a randomly selected bag of sugar. We are given that $X \sim N(\mu = 1000, \sigma^2 = 5^2)$. We want to find $P(990 < X < 1010)$. First, we convert the weights to z-scores: For $X = 990$, $z = \frac{990 - 1000}{5} = \frac{-10}{5} = -2$. For $X = 1010$, $z = \frac{1010 - 1000}{5} = \frac{10}{5} = 2$. So we need to find $P(-2 < Z < 2)$, where Z is the standard normal distribution. This probability can be found using a standard normal distribution table or calculator. The area to the left of $z = 2$ is.

© ClipSAT — clipsat.org | All Rights Reserved

44 **[Challenging]** • **Target: Variance-based reasoning with the Poisson distribution** • **Cross-Lesson: 6.03 Markov Chains (AI HL)**

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $P(X = 3) = P(X = 4)$, find the value of λ.

a. **4** ✓

b. 3

c. 12

d. $\frac{4}{3}$

For a random variable $X \sim Po(\lambda)$, the probability mass function is given by: $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$ Given $P(X = 3) = P(X = 4)$, we substitute $x = 3$ and $x = 4$: $\frac{e^{-\lambda}\lambda^3}{3!} = \frac{e^{-\lambda}\lambda^4}{4!}$ Since $e^{-\lambda} > 0$ and $\lambda > 0$, we can divide both sides by $e^{-\lambda}\lambda^3$: $\frac{1}{3!} = \frac{\lambda}{4!}$ Multiplyi.

45 **[Challenging]** • **Target: Combining Poisson reasoning with real-world decisions** • **Cross-Lesson: 1.01 Proof by Mathematical Induction**

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,P(X = 2) = P(X = 4)$, find the value of λ.

a. 6 ✓

b. 36

c. 12

d. $\sqrt{6}$

For a Poisson distribution $X \sim Po(\lambda)$, the probability mass function is given by: $P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}$ Substitute $r = 2$ and $r = 4$ into the given equation $3\,P(X = 2) = P(X = 4)$: $3\left(\frac{e^{-\lambda}\lambda^2}{2!}\right) = \frac{e^{-\lambda}\lambda^4}{4!}$ Since $e^{-\lambda} > 0$ and $\lambda > 0$, divide both sides by $e^{-\lambda}\lambda^2$: $\frac{3}{2} = \frac{\lambda^2}{24}$ Multiply both sides by 24: $\lambda^2 = 36$ Since.

© ClipSAT — clipsat.org | All Rights Reserved